Problem#5.1 from the Textbook

Define the conductor data

$GMR_{ACSR} := 0.0244\ ft$ $\quad R_{ACSR} := 0.306\ \frac{ohm}{mile}$ $\quad RD_{ACSR} := 0.03004\ ft$

$GMR_{ACSR_N} := 0.00814\ ft$ $\quad R_{ACSR_N} := 0.5920\ \frac{ohm}{mile}$ $\quad RD_{ACSR_N} := 0.02346\ ft$

$d_1 := 29i\ ft$ $\quad d_2 := (2.5 + 29i)\ ft$ $\quad d_3 := (7 + 29i)\ ft$ $\quad d_4 := (3 + 25i)\ ft$

$i := 1..4$ $\quad j := 1..4$ $\quad n := 1..3$ $\quad m := 1..3$

$S_{11} := \left|d_1 - \overline{d_1}\right| = 58\ ft$

$S_{12} := \left|d_1 - \overline{d_2}\right| = 58.054\ ft$

Define the image spacing matrix

$S_{i,j} := \left|d_i - \overline{d_j}\right|$

$$S = \begin{bmatrix} 58 & 58.054 & 58.421 & 54.083 \\ 58.054 & 58 & 58.174 & 54.002 \\ 58.421 & 58.174 & 58 & 54.148 \\ 54.083 & 54.002 & 54.148 & 50 \end{bmatrix} ft$$

Define the spacing matrix

$D_{i,j} := \left|d_i - d_j\right|$ $\quad D_{n,n} := RD_{ACSR}$ $\quad D_{4,4} := RD_{ACSR_N}$

$$D = \begin{bmatrix} 0.03 & 2.5 & 7 & 5 \\ 2.5 & 0.03 & 4.5 & 4.031 \\ 7 & 4.5 & 0.03 & 5.657 \\ 5 & 4.031 & 5.657 & 0.023 \end{bmatrix} ft$$

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$$P_{i,j} := 11.17689 \cdot \ln\left(\frac{S_{i,j}}{D_{i,j}}\right) \frac{\text{mile}}{\mu\text{F}}$$

Compute the Potential Coefficient Matrix

$$P = \begin{bmatrix} 84.561 & 35.152 & 23.715 & 26.613 \\ 35.152 & 84.561 & 28.606 & 29.004 \\ 23.715 & 28.606 & 84.561 & 25.247 \\ 26.613 & 29.004 & 25.247 & 85.665 \end{bmatrix} \frac{\text{mile}}{\mu\text{F}}$$

$$P_{ij} := \text{submatrix}(P, 1, 3, 1, 3) = \begin{bmatrix} 84.561 & 35.152 & 23.715 \\ 35.152 & 84.561 & 28.606 \\ 23.715 & 28.606 & 84.561 \end{bmatrix} \frac{\text{m}}{\mu}$$

$$P_{in} := \text{submatrix}(P, 1, 3, 4, 4) = \begin{bmatrix} 26.613 \\ 29.004 \\ 25.247 \end{bmatrix} \frac{\text{mile}}{\mu\text{F}}$$

$$P_{nj} := \text{submatrix}(P, 4, 4, 1, 3) = \begin{bmatrix} 26.613 & 29.004 & 25.247 \end{bmatrix}$$

$$P_{nn} := \text{submatrix}(P, 4, 4, 4, 4) = \begin{bmatrix} 85.665 \end{bmatrix} \frac{\text{mile}}{\mu\text{F}}$$

Apply Kron Reduction

$$P_{abc} := P_{ij} - \left(P_{in} \cdot P_{nn}^{-1} \cdot P_{nj}\right) = \begin{bmatrix} 76.293 & 26.142 & 15.871 \\ 26.142 & 74.741 & 20.058 \\ 15.871 & 20.058 & 77.12 \end{bmatrix} \frac{\text{mile}}{\mu\text{F}}$$

Calculate the Capacitance Matrix

$$C_{abc} := P_{abc}^{-1}$$

$$C_{abc} = \begin{bmatrix} 0.0151 & -0.0048 & -0.0019 \\ -0.0048 & 0.0159 & -0.0031 \\ -0.0019 & -0.0031 & 0.0142 \end{bmatrix} \frac{\mu\text{F}}{\text{mile}}$$

Calculate the Shunt Admittance Matrix

$$\text{Sei} := 10^{-6}\ \text{S}$$

$$f := 60\ \text{Hz} \qquad pi := 3.14\ \text{rad} \qquad \omega := 2 \cdot pi \cdot f$$

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$$Yabc := 1j \cdot \omega \cdot C_{abc}$$

$$Yabc = \begin{bmatrix} 5.704i & -1.806i & -0.704i \\ -1.806i & 5.992i & -1.187i \\ -0.704i & -1.187i & 5.339i \end{bmatrix} \frac{S \cdot 10^{-6}}{mile}$$

Calculate the Sequence Admittance Matrix

$$a := 1\angle 120\ deg$$

Define the sequence matrix, A_S

$$A_S := \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \qquad A_S = \begin{bmatrix} 1\angle 0^\circ & 1\angle 0^\circ & 1\angle 0^\circ \\ 1\angle 0^\circ & 1\angle -120^\circ & 1\angle 120^\circ \\ 1\angle 0^\circ & 1\angle 120^\circ & 1\angle -120^\circ \end{bmatrix}$$

$$Y_{012} := A_S^{-1} \cdot Yabc \cdot A_S$$

$$Y_{012} = \begin{bmatrix} 3.758 \cdot 10^{-18} + 3.214i & -0.13 - 0.01i & 0.13 - 0.01i \\ 0.13 - 0.01i & -8.32 \cdot 10^{-16} + 6.911i & 0.824 + 0.059i \\ -0.13 - 0.01i & -0.824 + 0.059i & 6.911i \end{bmatrix} \frac{S \cdot 10^{-6}}{mile}$$

$$a := 1\angle 120\ deg$$

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